Sweatshop on P-Chess: lane `core`, early morning

Input modes, premoves, history, review screen, openings, Lc0 failures and the flag race (10 tickets); GPT-6.1 Sol lineup

Foreman, session `sweatshop/core/2026-10-25-0159`

25 October 2026

Summary

Lane `core` ran in the clone `.artifacts-build/lanes/core`, from `main` at `5ee5836` (PRs #8 and #9 merged). Bruno authorized the relaunch with the `gpt-6.1-sol` lineup ("Pode relançar a lane core com a lineup gpt-6.1-sol"): one launch at 01:59, `-Models 'stage 2 gpt-6.1-sol high, hard gpt-6.1-sol xhigh, stage 3 gpt-6.1-sol xhigh'` (logged by the driver as `ad hoc`; the same models as the `Models:` binding), pid 21760, all stages on Codex. No `STOP` was asked for; the driver ended at 06:12 with "Nothing runnable". The standup expected five ready tickets (UI-006, UI-007, HIST-001, OPEN-001, BOT-008); each merge unblocked the next, so the run also took HIST-002, OPEN-002, OPEN-003, OPEN-004, and CLOCK-002 once Bruno chose its test seam mid-run.

- **Produced: 10 tickets approved and merged on the session** PR #10: UI-006, UI-007, HIST-001, HIST-002, OPEN-001, OPEN-002, OPEN-003, OPEN-004, BOT-008, CLOCK-002, all `Review: agent`, `Verdict: Approve`, no *Needs your call*. Gate at the last merge: 235 passed, 0 failed (from 152 before the run).
- **Cost: $25.44** at API list price over 28 stages, all priced: `gpt-6.1-sol xhigh` $15.52 ($12.11 in review, $3.41 as hard implementer), `gpt-6.1-sol high` $9.92. Codex quota: 26 points of the weekly window (`gpt-6.1-sol xhigh` 13, about 10 in review and 3 as hard implementer; `gpt-6.1-sol high` 11; 2 outside the stages). Section Codex quota has the detail.
- **What went wrong:** four tickets came back once each (UI-007, HIST-001, OPEN-001, OPEN-003), every one for a real miss against an explicit criterion; the four reopen rounds cost $5.95, 23% of the run. Nothing crashed, nothing was parked, and the foreman made no cleanup.
- **Provisional verdict:** the cheapest run of the project per ticket, $2.54 per approved ticket against $7.20–$8.30 for the Opus core runs. Reviews were short (median 5 min) and still caught each reopen with a reproduction. Manual product acceptance is pending on most tickets, and one known defect (Maia moving at zero, CLEAN-005) remains a release condition.

What was done

UI-006: the board takes moves by clicking a piece and then a square (an option, off by default), auto-queens promotions (off / on / only on premoves; `Alt` opens the picker anyway), and draws arrows with a right-drag and circles with a right click; a left click clears them. Mostly `ui/board_view.gd`; the shared promotion rule is `PromotionPicker.should_auto_queen`. Drawing while Maia is to move was handed to UI-007.

UI-007: premoves. While Maia thinks the player can queue moves, shown as ghost pieces; Lichess style holds one, Chess.com style a queue; each due premove plays at once, costs 0.1 s and plays its sound; an illegal one drops the rest; a right click cancels. New `ui/premove_queue.gd`. Reopened once: the ghost check let a queued move capture Black's king (`e2e4`, then `f1c4 d1h5 h5f7 f7e8`), because the queue replayed the human's moves with the ordinary generator, which never expects a king to be capturable. The real game was safe (the due move is checked again); the queue now refuses king captures.

HIST-001: every finished game is saved as a PGN with the clock after each move, and results are counted per Maia level and time control. New `core/history.gd`. Reopened once: the settings-screen test started the real game scene without a private history folder, so the new start-up code read the player's own `user://stats.cfg`; the test now gets an isolated history.

HIST-002: when a game ends, a review screen shows the result and the time spent on every move, and exports the game as PGN or opens it on Lichess, with rematch and back to setup. New `ui/review.gd`. The first move of each side shows 0 ms (a first premove 100 ms), following CLOCK-001's untimed first move.

OPEN-001: the full Lichess opening catalog (3,865 lines, CC0, pinned upstream commit) loads into a searchable tree with checkboxes, and selections can be saved by name for the Catalog mode. `bot/book.gd`, `ui/opening_picker.gd`. Reopened once: search ran on the tree's rebuilt names, in which `bot/book.gd` had rewritten the punctuation, so typing an original name with a comma or colon found nothing (46 lines, e.g. "Polish Opening, with d5"). The first catalog load takes about 17 s on a background thread; later openings reuse it.

OPEN-002: Maia now prefers the checked catalog lines early in the game and fades back to her own play later, with a configurable intensity; positions reached by transposition count; practice counts persist and saved games get an `Opening` header. `bot/book.gd`, `bot/maia\_player.gd`. Five scripted games against the real Lc0 confirmed the early preference and the fade.

OPEN-003: a player's repertoire can be imported from a PGN (Lichess study chapters, variations, comments), merged into one move tree and played in "My repertoire" mode with the saved colour (rematches swap colours). New `bot/repertoire.gd` and an import dialog. Reopened once: comments written before a chapter's first move were dropped from the merged tree.

OPEN-004: the review screen marks book moves, repertoire moves and the move where the player left the repertoire, with the expected continuation in SAN; the check can be switched off in every mode. New `bot/badges.gd`.

BOT-008: if Lc0 dies or rejects the Maia net, the game stops and returns to the setup screen with the error, instead of waiting forever. `engine/uci.gd`; an `error` line from Lc0 now counts as fatal even after `readyok`, because Lc0 v0.32.1 rejects a missing net and keeps running.

CLOCK-002: a piece dropped in the same frame the player's clock reached zero no longer plays; the clock is ticked before the board handles input (`ui/main.gd`, seven lines), using the test seam Bruno chose (the real game scene under the test runner, with a fake time source).

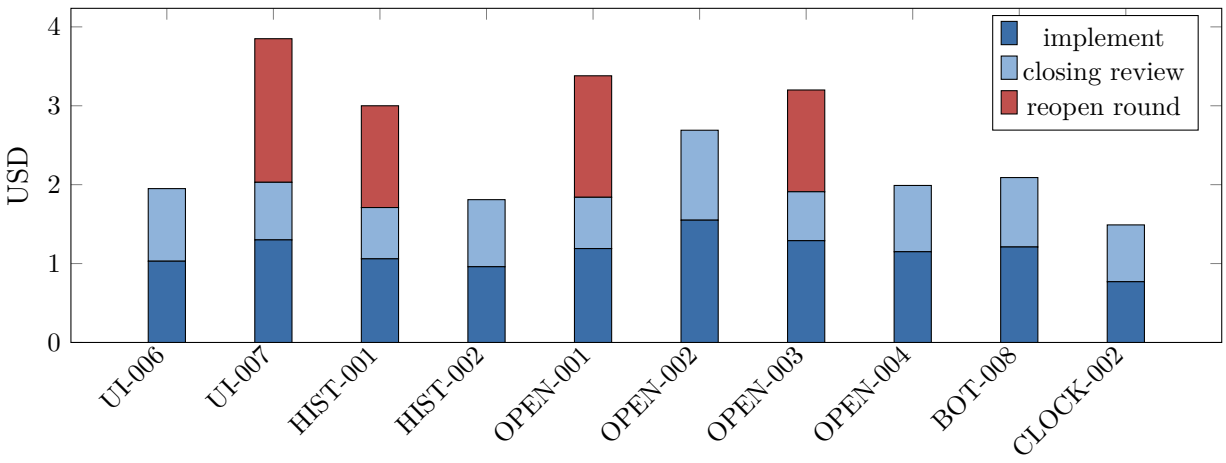
Timeline

01:59	Clone fast-forwarded to `5ee5836`; standup: five ready, CLOCK-002 blocked, tree clean, no driver. Launch, pid 21760, session `sweatshop/core/2026-10-25-0159`.
02:18	UI-006 merged, 0 findings. Bruno (relayed) unblocks CLOCK-002 with seam (a); the foreman commits the contract on the session by plumbing during UI-007's stage 2 (`b206a36`).
02:45	UI-007 reopened (king capture in the ghost). 02:58 merged.
03:13	HIST-001 reopened (settings test reads the player's stats). 03:22 merged.
03:38	HIST-002 merged, 0 findings.
04:04	OPEN-001 reopened (search loses punctuation). 04:15 merged.
04:42	OPEN-002 (hard) merged, 0 findings.
05:04	OPEN-003 reopened (chapter comments dropped). 05:11 merged.
05:33	OPEN-004 merged, 0 findings. 05:54 BOT-008 merged, 0 findings.
06:12	CLOCK-002 merged, 0 findings; "Nothing runnable"; PR #10 updated (10 tickets).

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
02:13	UI-006	impl	sol high	12	2.8M	95%	21k	1.03	to-review
02:18	UI-006	review	sol xhigh	5	2.1M	90%	13k	0.92	merged
02:40	UI-007	impl	sol xhigh	22	4.0M	98%	34k	1.30	to-review
02:45	UI-007	review	sol xhigh	6	3.0M	91%	16k	1.26	reopened
02:52	UI-007	impl	sol xhigh	6	1.6M	95%	10k	0.56	to-review
02:58	UI-007	review	sol xhigh	6	1.9M	93%	13k	0.73	merged
03:09	HIST-001	impl	sol high	11	2.9M	95%	22k	1.06	to-review
03:13	HIST-001	review	sol xhigh	4	1.8M	90%	10k	0.79	reopened
03:18	HIST-001	impl	sol high	4	1.2M	91%	8k	0.50	to-review
03:22	HIST-001	review	sol xhigh	5	1.5M	91%	11k	0.65	merged
03:34	HIST-002	impl	sol high	11	2.4M	93%	21k	0.96	to-review
03:38	HIST-002	review	sol xhigh	5	2.1M	92%	13k	0.85	merged
03:57	OPEN-001	impl	sol high	18	3.6M	98%	30k	1.19	to-review
04:04	OPEN-001	review	sol xhigh	8	3.0M	94%	16k	1.08	reopened
04:10	OPEN-001	impl	sol high	6	1.3M	94%	7k	0.47	to-review
04:15	OPEN-001	review	sol xhigh	5	1.7M	93%	10k	0.65	merged
04:36	OPEN-002	impl	sol xhigh	21	4.0M	95%	37k	1.55	to-review
04:42	OPEN-002	review	sol xhigh	6	3.1M	93%	15k	1.14	merged
04:58	OPEN-003	impl	sol high	16	3.7M	97%	32k	1.29	to-review
05:04	OPEN-003	review	sol xhigh	5	2.5M	92%	13k	0.99	reopened
05:07	OPEN-003	impl	sol high	3	0.8M	94%	5k	0.30	to-review
05:11	OPEN-003	review	sol xhigh	4	1.4M	90%	10k	0.62	merged
05:27	OPEN-004	impl	sol high	16	3.0M	95%	29k	1.15	to-review
05:33	OPEN-004	review	sol xhigh	6	2.1M	92%	12k	0.84	merged
05:50	BOT-008	impl	sol high	17	3.7M	96%	22k	1.21	to-review
05:54	BOT-008	review	sol xhigh	4	2.2M	91%	11k	0.88	merged
06:04	CLOCK-002	impl	sol high	10	2.3M	96%	14k	0.77	to-review
06:12	CLOCK-002	review	sol xhigh	7	1.8M	93%	12k	0.72	merged
				249	67.5M		467k	25.44	

"sol" is `gpt-6.1-sol`. UI-007 and OPEN-002 ran on the hard implementer (`Difficulty: hard`).



The red segments are the four reopen rounds (the review that sent the ticket back plus the retry), $5.95 in all. Every retry was small (3–6 min, $0.30–$0.56): the findings were narrow and came with a reproduction. Input is 90–98% cache reads; the costliest stage was OPEN-002's hard implementation ($1.55).

Comparison with previous runs

From `run-log.md`: lane `core` on Opus cost $7.20 per approved ticket (night) and $8.30 (afternoon); the UI-004/UI-005 runs and their reworks $23.01 for two tickets; lane `mvp` on this same lineup $2.69. This run: $2.54 ($25.44 over 10). Stage 2 took 3–22 min (median 11.5) and reviews 4–8 min (median 5), against 12–29 min for Opus high on the afternoon run. The reopen rate was 4 of 14 reviews (29%), above the lineup's history (16% before this run); every reopen was a single narrow finding.

Codex quota

From `codex-quota.ps1` over the run's window (01:59–06:13), baseline 25% at 01:54. The total is exact; one stage is ±1 point, since Codex rounds to integers. The script flags a weekly-window boundary during UI-007's first review, so that stage's point may be misread.

Model and role	Cost (USD)	Weekly quota
`gpt-6.1-sol high`, stage 2	9.92	11 points
`gpt-6.1-sol xhigh`, stage 3	12.11	about 10 points
`gpt-6.1-sol xhigh`, hard stage 2 (UI-007, OPEN-002)	3.41	about 3 points
Outside the stages (8 threads)	–	2 points
Run total	25.44	26 points

The 8 threads outside the stages are other Codex work in the same account during the window (the stages' `gpt-6-astra` proxy calls run inside their stage).

Observations per model

- **gpt-6.1-sol high, stage 2 (8 tickets, 11 runs)**: test first in every cycle, with separate test and production commits and red results recorded; every handoff named what it did not check (visual, audio, clipboard, browser). It asked the proxy whenever a fix needed a file outside Primary files, instead of editing it. Its three reopens were each a path it did not walk: one more caller of the new start-up code (HIST-001), searching the names it had itself rewritten (OPEN-001), and a comment position in its own parser's output (OPEN-003).
- **gpt-6.1-sol xhigh, hard stage 2 (UI-007, OPEN-002)**: the two largest tickets, 21–22 min each. UI-007 reconciled UI-006's handoff (drawing during Maia's turn) without being reminded; it missed the king-capture case in the ghost board. OPEN-002 passed review at once, with five real-engine games as acceptance.
- **gpt-6.1-sol xhigh, review (14 reviews)**: 10 merges, 4 reopens, each with a reproduction and the exact lines; no code authored in review, so no round-0 rows. It filed five follow-up tickets for things outside the criteria (CLEAN-002 to CLEAN-006) instead of leaving notes in closed tickets, which keeps them visible.

Driver and runtime issues

1. **None blocking.** No crash, no park, no failed attempt; every stage ran on Codex and produced its gate receipt with `gate-evidence.ps1`.
2. **A ticket unblocked mid-run.** Bruno chose CLOCK-002's seam while the driver ran. The driver reads tickets from the session ref at every loop, so the foreman committed the contract with Git plumbing (`commit-tree` and a compare-and-swap `update-ref`) while the clone sat on UI-007's ticket branch, and pushed it. The driver picked CLOCK-002 up after BOT-008 without a relaunch.

3. **Scene tests multiply.** Five tickets added start-up work to the real game scene, and three of them had to adapt other tickets' scene-test helpers (HIST-001, OPEN-002, OPEN-003), each by a proxy decision. The gate grew from 152 to 235 tests.

Recommendations

1. Keep the gpt-6.1-sol lineup for lane `core`: $2.54 per ticket. Scoreboard rows: `gpt-6.1-sol high` → `gpt-6.1-sol xhigh` 102 reviews, 16% reopened, S2 rate 15%; `opus-5.5 high` → `opus-5.5 xhigh` 12 reviews, 17% (a small sample, at about three times the cost per ticket).
2. Stage 1 should triage the five CLEAN tickets the reviews filed. CLEAN-005 (Maia's move can land at zero) is a release condition; CLEAN-006 holds the real-Main rematch tests UI-004 still owes.
3. When a ticket adds start-up code to the main scene, its stage 2 should grep every test that instantiates `ui/main.tscn`. HIST-001's reopen came from the one caller it missed, and OPEN-002 then had to touch four scene-test helpers.
4. The 17 s first catalog load is not under any criterion. If it bothers Bruno in play, a cached parse of the five TSVs is a small follow-up.

Human debt and proxy decisions

No `Débito humano:` lines this run. Bruno decided twice, relayed by the supervising session: relaunch lane `core` on the gpt-6.1-sol lineup; and CLOCK-002's test seam, option (a) ("pode destravar o clock-002 também, aceito sua sugestão"). Every `Proxy decided` line of the run is `(gpt-6-astra high, fallback)`, 12 in all:

- **UI-006**: add the two Main settings routes and criterion-focused tests; retain the existing turn-input lock for UI-006, leaving opponent-turn input integration with UI-007.
- **UI-007** (written during review): reopen for criterion 2; queued moves must not capture the opposing king in the ghost position; unknown Maia replies are handled by execution-time revalidation and do not excuse accepting an invalid move.
- **HIST-001**: isolated History harnesses in `test_session.gd` and `test_premove.gd` plus the real-Main history integration seam.
- **HIST-001**: isolated History injection, storage lifecycle and a start-up identity assertion in `test_input_modes.gd`, preserving its settings assertions.
- **HIST-002**: criterion 2 reconstructs charged time using only awarded increments, preserving 100 ms first premoves and zero-clock snapshots without increment; scene tests in `tests/test_review.gd`.
- **OPEN-002**: narrowly scoped Main lifecycle wiring and scene tests in `test_book.gd`; criterion 7 clarified to allow positional re-entry.
- **OPEN-002**: `opening_picker.gd` gets one shared public asynchronous catalog-readiness entry point used by Main and `present()`.
- **OPEN-002**: tiny preloaded-catalog injection in the `_session` helpers of four existing scene-test files, assertions unchanged.
- **OPEN-003**: Main repertoire lifecycle wiring and real-scene tests; the imported colour for setup starts, the actual game colour for rematches.
- **OPEN-003**: update only OPEN-001's temporary "My repertoire disabled" assertion to expect the mode enabled.
- **OPEN-004**: Main composition wiring; the Book badge independent of the repertoire toggle; only the first unmatched human move marked, with explicit no-continuation text.
- **BOT-008**: fail closed on Lc0's `error` output even after `readyok`; exercise missing weights through the next policy request.

Feature acceptance

Against the milestones in `docs/spec.md`, reusing the stages' evidence on identical code:

- **M5** (every setting works and persists; UI-005, UI-006, UI-007): automated persistence and input tests pass; *pending* the manual checks (gesture feel, ghosts, premove audio).
- **M6** (badges and times correct on saved games; HIST-001, HIST-002, with OPEN-004's badges): automated real-Main games and saved-PGN checks pass; *pending* two interactive games, the real clipboard and the Lichess import.
- **M7** (Maia follows the chosen lines and drifts off them; OPEN-001 to OPEN-004): *passed* on automated evidence, five scripted games against the real Lc0 (OPEN-002) and real Maia-1500 repertoire play (OPEN-003); the visual check is pending.
- **M4** addition (CLOCK-002): *passed* headless through real viewport input. Known deferred defect: Maia's own move can still land at zero (CLEAN-005), a release condition.

Reopen attribution

Four reopened reviews, one finding each; judged by the foreman (Opus), since the reviewer is `gpt-6.1-sol`. No merging review authored a fix, so no round-0 rows. PR #10 has no review findings yet; no session-review, release-audit or post-release discovery this run.

Ticket	Rnd	Finding	Label	Why
UI-007	1	Ghost validation accepted capturing the opposing king	S2-explícito/código	Criterion 2: each premove checked against the position after the queued ones.
HIST-001	1	Settings test's real Main read the player's `stats.cfg`	S2-explícito/código	Criterion 7: every automated Main test uses isolated history; a regression of stage 2's own startup code.
OPEN-001	1	Search lost original punctuation (46 lines)	S2-explícito/código ?	Criterion 4: substring search; borderline as the criterion does not say original or displayed names.
OPEN-003	1	Chapter-leading comments not merged into the tree	S2-explícito/código	Criterion 1: every chapter's comments merge into one tree.

Totals: S2-explícito/código 4 (one borderline). Review miss: none (all first reviews). Churn: none.

Scoreboard

`scoreboard.ps1` over P-Chess, ClassyCall, EasyMath, PhysiSyst, SIGAA-ME, SynchroNice and slip. For P-Chess the lane clone was read instead of the main checkout, after the merged log was copied back to both: the clone also holds this run's uncommitted attribution and intervention rows, and one P-Chess path is read, so no row counts twice. Rows that moved since the previous `core` report: `gpt-6.1-sol high` → `gpt-6.1-sol xhigh` 89 → 102 reviews, 12 → 16 reopened (13% → 16%), S2 11 → 15; a new row `gpt-6.1-sol xhigh` → `gpt-6.1-sol xhigh`, 4 reviews, 1 reopened (an anecdote); `opus-5.5 high` → `opus-5.5 xhigh` 8 → 12 (the UI-004/UI-005 runs).

Implementer	Reviewer	Reviews	Reopened	Rate	S2	S2 rate	Fixed in rev.
gpt-6.1-sol high	gpt-6.1-sol xhigh	102	16	16%	15	15%	6
gpt-6.1-sol high	gpt-6.1-sol max	94	21	22%	20	21%	4
gpt-6.1-sol xhigh	gpt-6.1-sol xhigh	4	1	25%	1	25%	1
opus-5.5 high	opus-5.5 xhigh	12	2	17%	2	17%	4
opus-5.5 xhigh	opus-5.5 xhigh	8	0	0%	0	0%	0
?	opus-5.5 xhigh	4	1	25%	1	25%	0

Implementers: `gpt-6.1-sol high` 229 runs, 200 to review, 8 failed, 21 asked, median 19 min, $380.59; `gpt-6.1-sol xhigh` 4 runs, all to review, $4.74. Discoveries: 252 ticket-review findings,

13 session-review (9 of 10 known were earlier review misses). Interventions across repos: 24 human rescues, 61 automatic recoveries, 14 routine approvals, 44 `gpt-6-astra high, fallback`, 29 Opus and 12 supervisor proxy decisions, 37 unknown. Runtime completion to process exit: 917 s over 669 measured stages, 763 unmeasured.

Pending for the human

1. Review and merge PR #10 (10 tickets); then tell the other open lanes' foremen to sync.
2. Manual checks: UI-006 (click-click, promotion, arrows), UI-007 (a 1+0 game with premoves, ghosts, audio), HIST-001 (two games, then open their PGNs in `user://games/`), HIST-002 (copy the PGN, open on Lichess, compare), BOT-008 (kill Lc0 in Task Manager mid-game), plus the visual look of the opening picker, repertoire import and review badges.
3. Stage 1: triage CLEAN-002 to CLEAN-006; CLEAN-005 (Maia's move at zero) is a release condition.
4. Carried over: UI-004 and UI-005 manual checks, BOT-004's latency on an idle machine, CLOCK-001's manual games, `PolicyTemperature`.